0.5 Project Scope Sheet · Group 1 · Fernão de Oliveira

Draft for discussion · 12 November 2026 · T. Santos, L. Batista

Ship and Ocean Systems Design 2026 · Ship Design Project · Scope sheet for your group

Field	Value
Group	Group 1 · Fernão de Oliveira
Members (student number)	106238 · 107304 · 106645
Baseline	B · Azores Feeder: geared container ship, 130 TEU, battery-electric with swappable battery containers, two island loops plus a transit each week
Specialization 1	SP01 · Longitudinal Strength Calculation
Specialization 2	SP02 · Propulsion Matching
Deadline	Dec 13, 2026, one ZIP file by e-mail to luis@veranavis.com
ZIP name	Group1_Project.zip
Presentation	Dec 18, 2026, 15 minutes per group

Revisions

All alterations to the project scope of work and/or design requirements for the group, against 0.2 Ship Design Project · Baseline B, Azores Feeder: Owner's Requirements (p. 5), are recorded in the following table:

Rev.	Date	Description	By
0	To be set	First issue, after discussion with the students	T. Santos, L. Batista

Baseline B

Deliverable Groups D1–D8 and D10

ID	Deliverable	Status
D1	**Design Basis.** 130 TEU at 10 t per TEU (1,300 t cargo) · battery containers as extra deadweight, mass from supplier data · 11–12 kn · two loops plus the transit in a week · limits at all eight ports, each with its source · every requirement tagged Given / Assumed / To be checked against 0.2 Ship Design Project · Baseline B, Azores Feeder: Owner's Requirements (p. 5). **What is evaluated:** the transportation problem: ship type and target capacity, the sequence of port calls (round trip), the relevant characteristics of the ports	
STU-D1.1	Owner's requirements and port limits	
STU-D1.2	Laws and regulations applicable, incl. environmental targets	
D2	**Reference Ship Database.** At least 8 small geared container ships and battery ships · other families as your chosen hull needs · sources for every ship · your own regressions and bounds. **What is evaluated:** the ship database compiled and the analysis of its statistical data	
STU-D2.1	Reference ship database with sources	
STU-D2.2	Regressions and bounds	

ID	Deliverable	Status
D3	**Ship Model.** Inputs, outputs and calculation sequence (volume carrier) · hull form coefficients (C_B, C_P, C_M, C_WP, vertical prismatic) · tonnage (GT, NT) · lightship weight: steel, outfit, machinery and superstructure, crane included, with centres · deadweight components: cargo, batteries as deadweight, lubricating oil, fresh water, provisions and stores, crew · propulsive coefficients (wake, thrust deduction, relative-rotative and hull efficiency) · propulsive power by Holtrop & Mennen, electric drive, two shafts · propeller selection with a cavitation check · side thruster · electric power estimate for each operating condition · subdivision and compartments (bulkheads, engine room and peaks) · capacities of holds and tanks, containers in the holds and on deck · closed weight balance. **What is evaluated:** sequence and extent of the computations · selection of the estimation methods · correct response of the outputs to changes in the inputs (qualitative) · every item estimated, with the method named and its range of validity respected	
STU-D3.1	Ship model (computer codes and/or files)	
STU-D3.2	Weight estimation, assumptions and margins	
D4	**Voyage and Financial Model.** Both loops and the transit: energy per leg, battery swaps and charging, electricity and port costs · building cost, financing, RFR and NPV · ship model outputs as input. **What is evaluated:** times and costs at each port · times and energy consumption at sea · global results for one round trip and for one year of operation · financial analysis	
STU-D4.1	Voyage model	
STU-D4.2	Financial model	
D5	**Validation Note.** Your model against the reference families you chose, with the choice justified · error per quantity · model corrected where needed. **What is evaluated:** validation of the ship model: lightship weight, propulsive and electric generation power, newbuilding cost	
STU-D5.1	Validation against reference ships	
D6	**Optimization Report.** Minimum battery capacity that completes both loops in one week · justified objective · active constraints · optimum against the requirements · code flowchart: a flowchart of your code that shows how the information flows (inputs, each calculation step, the iterations and the outputs), in the symbols of ISO 5807. **What is evaluated:** problem correctly defined: design variables, restrictions (with their sources), objective(s) · exploration of the domain of the variables · the code delivered reproduces the results presented · the flowchart matches the code, shows the inputs, calculation steps, iterations and outputs, and uses the ISO 5807 symbols correctly	
STU-D6.1	Optimization report against the design objective, sensitivity analysis	
STU-D6.2	Code flowchart (ISO 5807)	
D7	**Compliance Sheet.** EEDI and CII: show why they do not apply, and report energy per TEU-mile · battery rules · stability while the crane lifts · freeboard · draught at every port. **What is evaluated:** each applicable rule identified with its source, and the result shown against the limit (freeboard, stability with the crane lifting, draught at every port) · EEDI and CII shown not to apply, with the reason	
STU-D7.1	EEDI and CII	
STU-D7.2	Preliminary intact stability assessment: requirements and typical conditions, incl. ballast conditions	
STU-D7.3	Draught at every port	
D8	**General Arrangement and Hull Form.** Deck container layout and crane reach · battery slots aft · container count on your own arrangement · 3D hull form, justified · general arrangement (profile, midship section and main deck), showing the container and battery layout and the crane reach · 3D hull shape with the containers in the holds and on deck, the main cargo-hold bulkheads, the collision bulkhead and the deckhouse exterior · hydrostatics check. **What is evaluated:** general arrangement with the main dimensions and the cargo area configuration · final main dimensions and hydrostatic properties (displacement, C_B) against the targets, with the error levels discussed · final cargo capacity (TEU, deadweight) against the targets, with the error level discussed	
STU-D8.1	Hydrostatics check	

ID	Deliverable	Status
DWG-D8.1	General arrangement (A3): profile, midship section and main deck, with the container and battery layout and the crane reach	
D10	**Datasheet and Presentation.** One-page A4 datasheet (0.4 Datasheet Template) · report on the 0.3 Report Template · oral presentation of the results in PowerPoint, no longer than 15 minutes, including: objectives of the work (description of the initial concept of the ship); computation model (the computation process, with the adopted methods, criteria and validations); assessment of the results obtained (comparison with existing similar ships); conclusions. **What is evaluated:** report: a clear description of all the work, following the 0.3 Report Template · summary of the resulting ship characteristics · comparison with the statistical data of existing ships of the same market segment · presentation as described in D10	
STU-D10.1	Datasheet (template 0.4)	
STU-D10.2	Presentation, 15 minutes	

Specialization scope

One calculation note per specialization goes in D9, with the drawings listed. Each specialization has its own self-study deck.

Deliverable Group D9

ID	Deliverable	Status
D9	**Specialization Documents (2).** One individual report per specialization, with its required studies and drawings (below). **What is evaluated:** every required study and drawing delivered · methods and assumptions stated · results checked against the acceptance criteria of the specialization	

SP01 · Longitudinal Strength Calculation

Show that the hull girder carries the still-water and wave bending moments.

ID	SP01 deliverable	Status
STU-SP01.1	Lightship weight and deadweight distribution along the length	
STU-SP01.2	Four critical loading conditions (subject to justification and/or definition with the instructor)	
STU-SP01.3	Still-water and wave bending moments and shear forces for each loading condition	
STU-SP01.4	Allowable stresses, in bending and in shear	
STU-SP01.5	Preliminary scantlings of the midship cross-section	
STU-SP01.6	Section modulus, moment of inertia and neutral axis of the midship section	
STU-SP01.7	Allowable vertical still-water bending moment and shear force: the allowable envelopes along the length, with each loading condition plotted against them	
DWG-SP01.1	Midship cross-section	

SP02 · Propulsion Matching

Choose the propeller and match it to the motors in every operating mode.

The SP02 report is a resistance and propulsion report, justifying:

ID	SP02 deliverable	Status
STU-SP02.1	Detailed ship resistance: each resistance component, the margins and the assumptions, computed from 4 to 14 knots in steps of 1 knot; comparison with the Baseline results. Updated	
STU-SP02.2	Design statement and assumptions; design, machinery and sea margins; efficiencies of the mechanical and electrical equipment	

ID	SP02 deliverable	Status
STU-SP02.3	Optimum Wageningen B-series propeller for the trial condition, the service speed, the top speed and the mainland transit (7 knots)	
STU-SP02.4	Selection of the propeller design condition, with justification; comparison with the Baseline results	
STU-SP02.5	Propeller model description and design features; typical E-motors identified (depending on the type, a gearbox might not be required)	
STU-SP02.6	Generator sizing: one generator able to run both electric motors at the mainland transit speed, with and without the sea margin; typical genset equipment identified	
STU-SP02.7	Engine matching: operating points on the E-motor layout diagram, with the sea margin	
STU-SP02.8	Assumptions on the methodology for the shore–ship electrical interface; battery sizing: calculations, assumptions and high-level procurement specifications	
DWG-SP02.1	Propulsion arrangement: propeller, shaft line (preliminary sizing) and rudder, with the equipment identified	